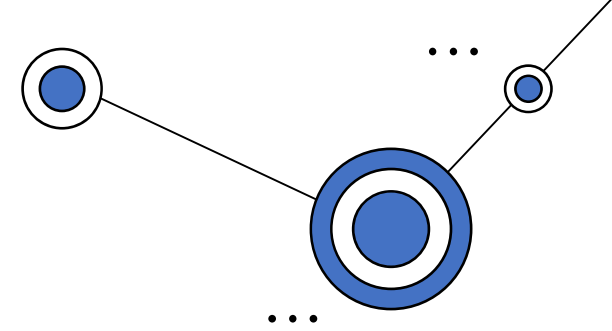
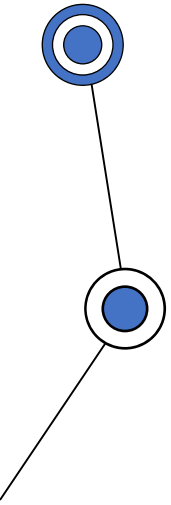
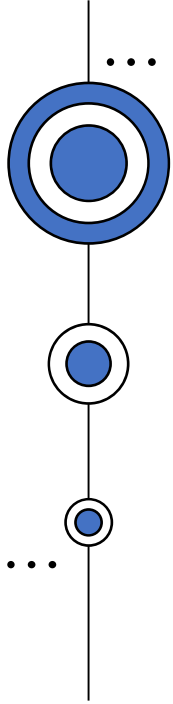




7 Revealing Ways AIs FAIL

Presenter: Shahzad Momayez
Winter 1403





In The Name of God

...

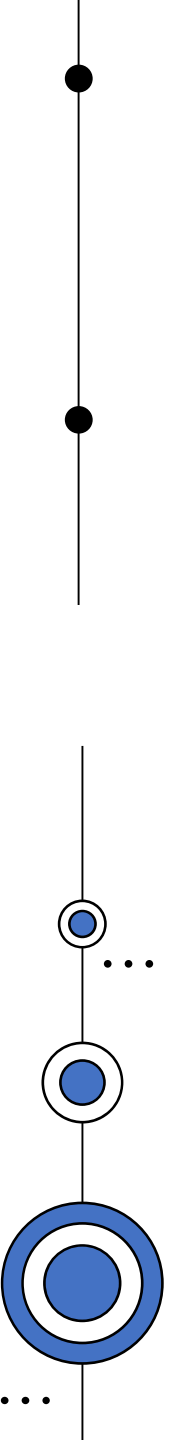
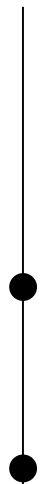


Table of Contents

00

Introduction

...

01

Brittleness

...

02

Embedded Bias

...

03

Catastrophic Forgetting

...

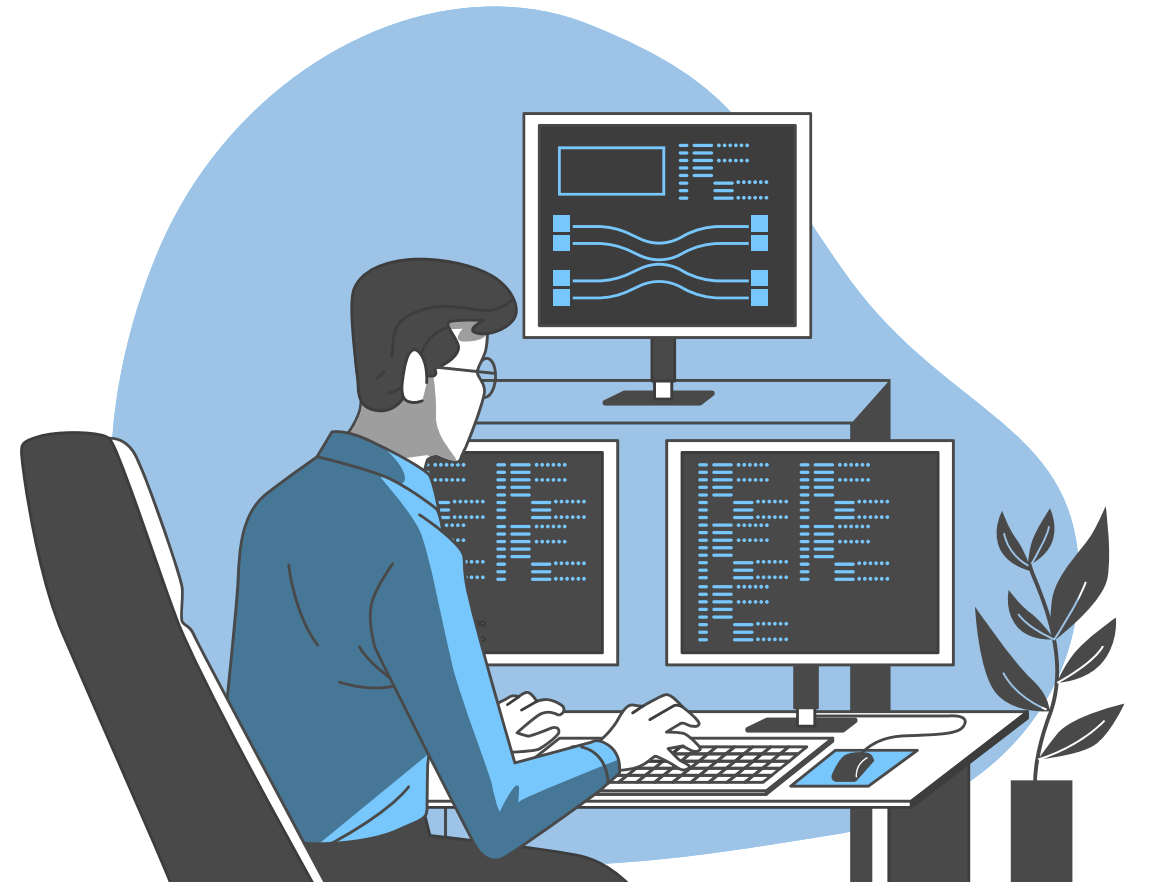


Table of Contents

04

Explainability

...

05

Quantifying Uncertainty

...

06

Common Sense

...

07

Math

...

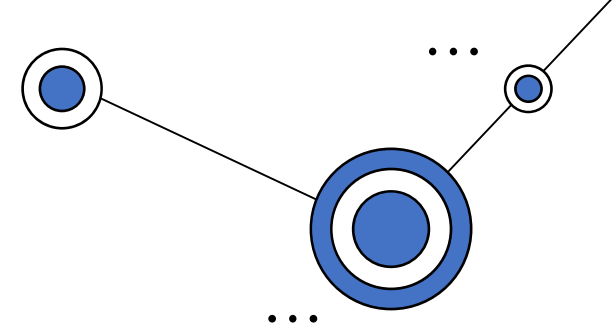
08

Conclusion

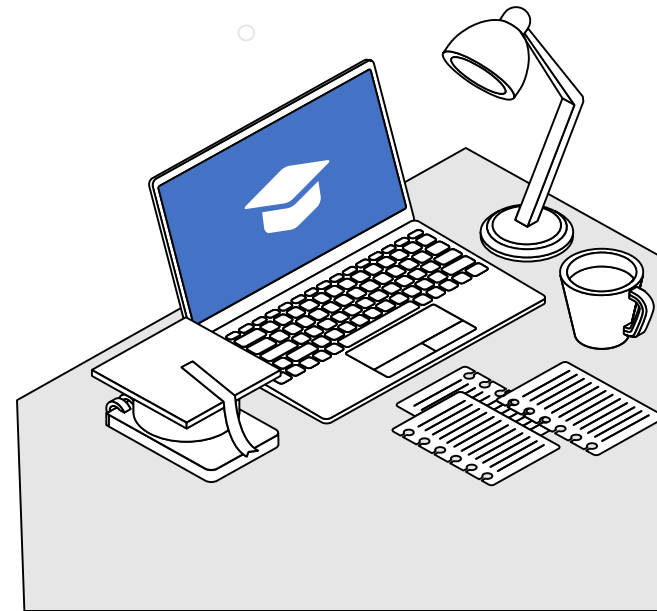
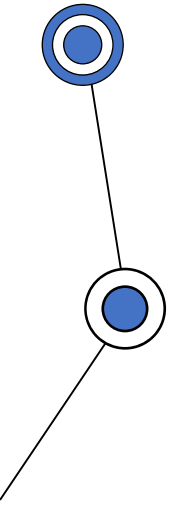
...



Introduction



The article explores seven critical weaknesses of artificial intelligence (AI), emphasizing their real-world implications and potential remedies.



Artificial intelligence (AI) is changing the world **by solving problems faster**, more **accurately**, and more **reliably** than humans.

It's used for tasks like:

1. **Detecting diseases**
2. **Helping in hiring decisions**
3. **Driving cars**
4. ...

But AI is far from perfect. It has serious flaws. AI fails is important so we can **fix these issues** and build better, more trustworthy systems.

This article explains seven major ways AI fails and what we are doing to address these problems.

**What are
these 7 flaws
of the AIs?**

01

Brittleness

...

AI systems often fail when faced with situations they haven't been trained for.

For example, an AI that could recognize a school bus struggled when the image of the bus was flipped on its side and it misidentified it as a snowplow.

Even small changes, like **putting stickers on stop signs** or **changing a single pixel** in an image, can confuse AI.

This happens because AI can only recognize patterns it has seen before.

Flipped



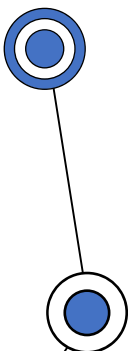
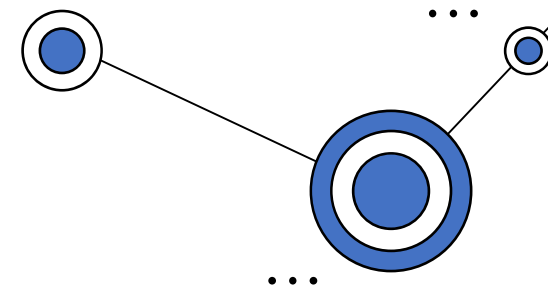
What is the label of this picture?

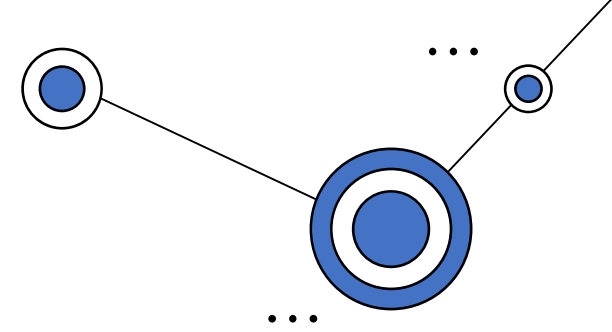


Snowplow



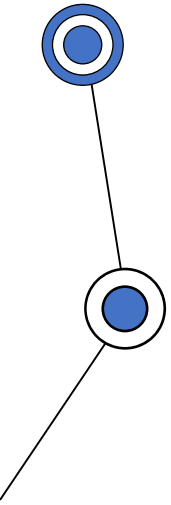
School Bus

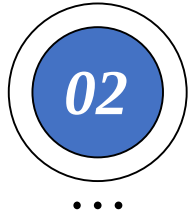




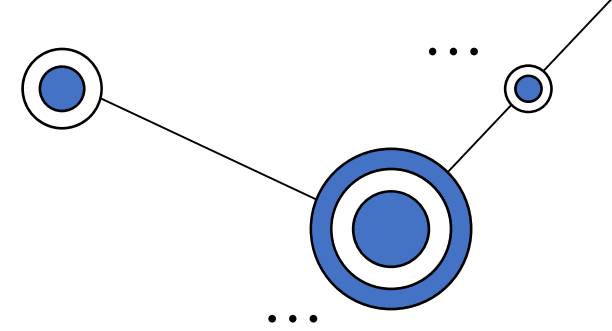
Remedies:

To make AI stronger, researchers train it using “**adversarial examples**” which are tricky scenarios designed to **confuse the system**. This helps AI learn **to handle unexpected changes**. However, rare, unpredictable events (like pandemics or natural disasters) still pose a challenge, even for humans.

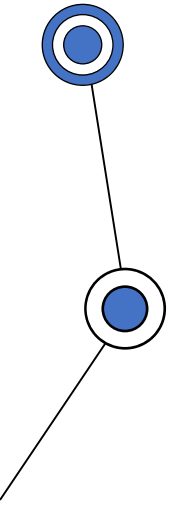


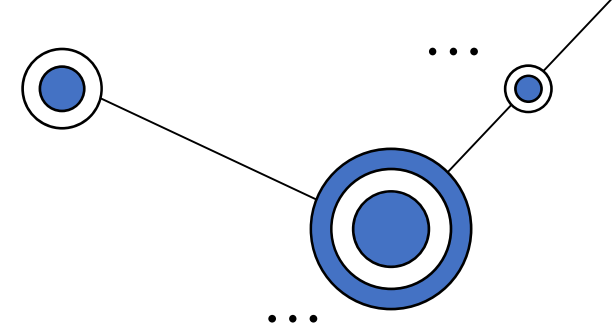


Embedded Bias



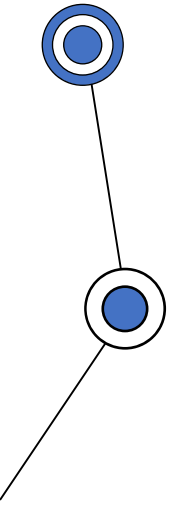
AI can pick up and amplify biases present in its training data. For instance, a healthcare AI prioritized **healthier white patients** over **sicker Black patients** because it assumed **higher medical costs** meant **more severe illness**. This overlooked systemic inequalities in healthcare access.

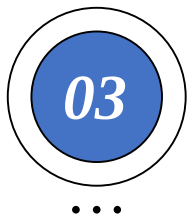




Remedies:

- Researchers suggest retraining AI **with better datasets** that **represent all groups fairly**. Regularly testing the AI for biased behavior and having oversight teams monitor its decisions can also help.
- **Developing algorithms** that focus **on fairness and transparency** is another effective solution.

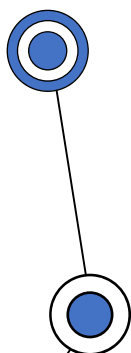
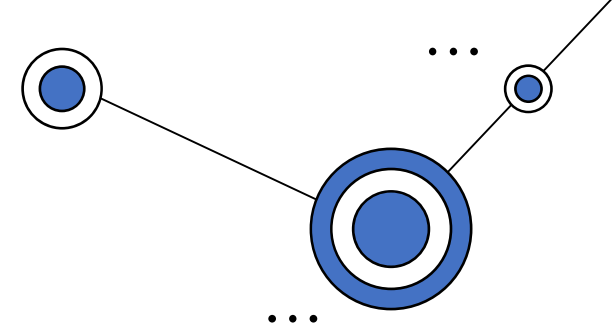


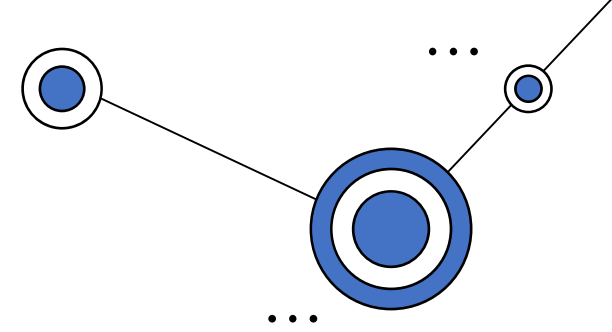


Catastrophic Forgetting

AI tends to forget what it learned previously when it's trained on new tasks.

For example, an AI detecting fakes stopped **recognizing older types of fake videos** after being updated to identify new ones.

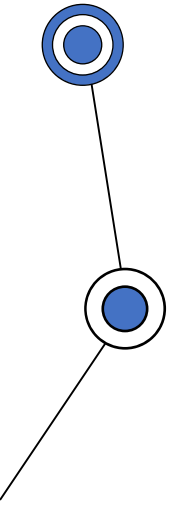




Remedies:

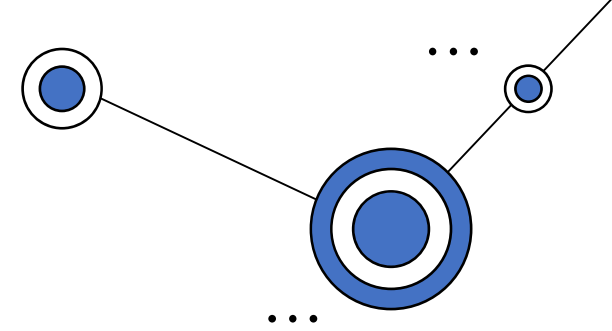
- One solution is “knowledge distillation”. It means that **older AI systems pass their knowledge to newer ones.**
- Another approach involves **periodically refreshing the AI with summaries of old data.**

Researchers are also developing continuous learning techniques that allow AI to retain knowledge without relying on **private** or **outdated** data.



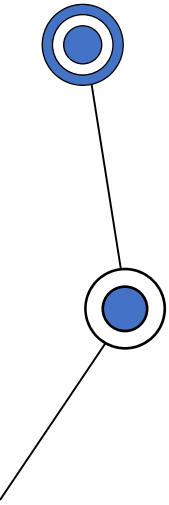


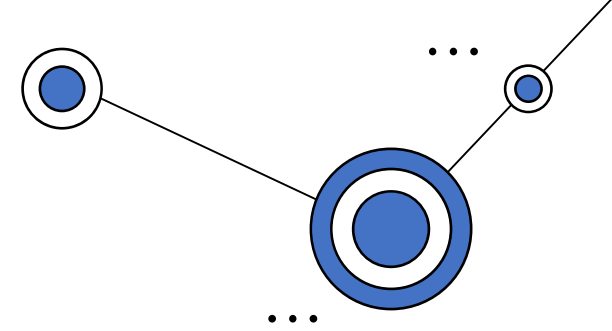
Explainability



AI decisions often feel like a “**black box**”. It means that people don’t know **why** or **how** an AI made a particular decision.

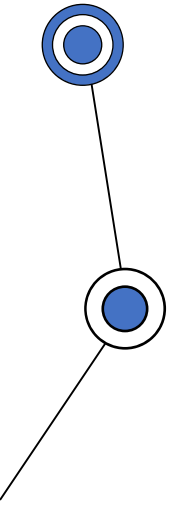
For example, it’s hard to understand why an AI suspects someone has **cancer** or **predicts a criminal risk score**.





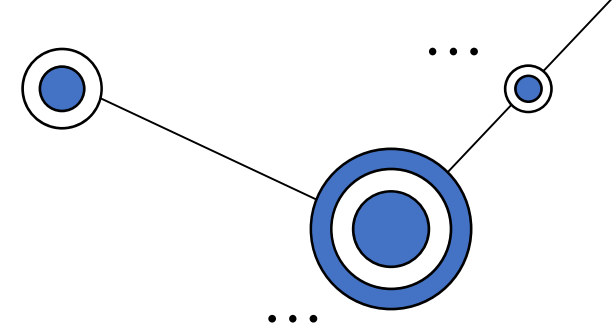
Remedies:

- Researchers are creating tools to explain AI decisions in simple, clear ways. One idea is to **build databases of correct explanations** that AI can reference.
- Another approach is to develop **standardized methods** that make AI behavior **more predictable and consistent**.



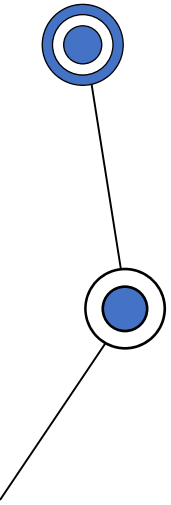


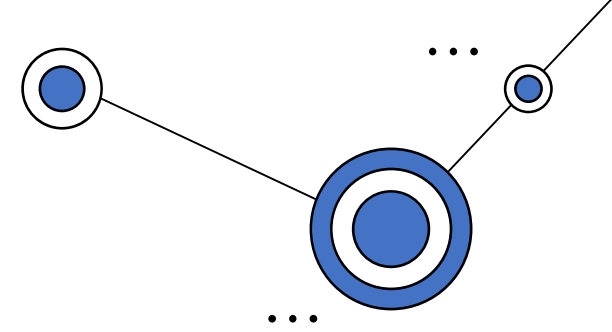
Quantifying Uncertainty



AI can **act overconfident** even when it's wrong.

- For instance, in 2016, a **Tesla autopilot** system failed to recognize a truck and caused a **fatal crash**. It didn't realize it was uncertain about the situation.

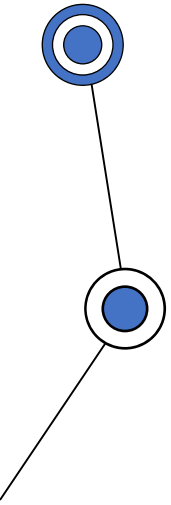


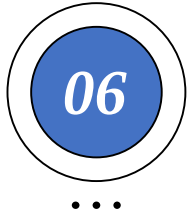


Remedies:

- AI systems can be **improved to measure their own uncertainty** and alert humans when unsure.
- For example, some AI systems in healthcare now **indicate confidence levels for diagnoses**, which helps doctors make better decisions.

Researchers are working on faster techniques to apply this idea **in real-time systems** like self-driving cars.



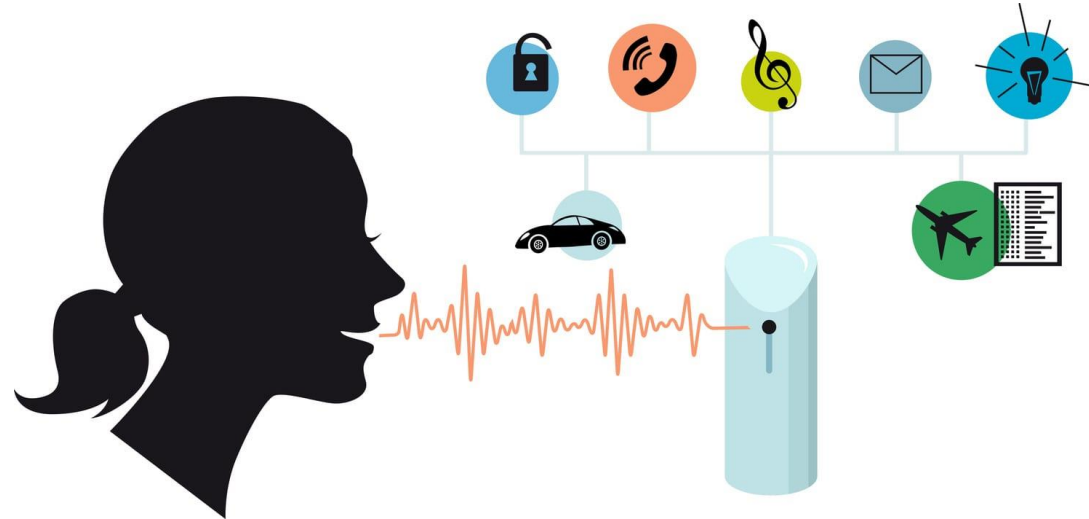


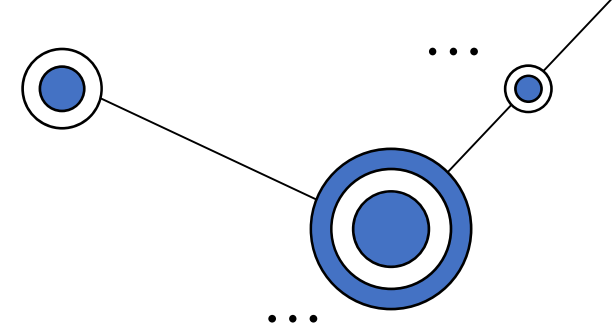
Common Sense

AI lacks common sense, which means it often fails to understand context.

For example, hate speech detection systems can mistakenly label **harmless phrases** as **offensive** because they don't grasp the intent or **tone** behind the words.

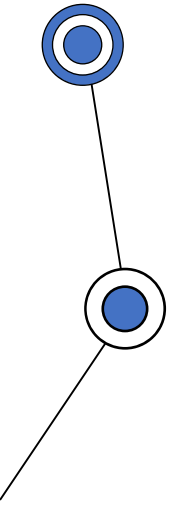
.





Remedies:

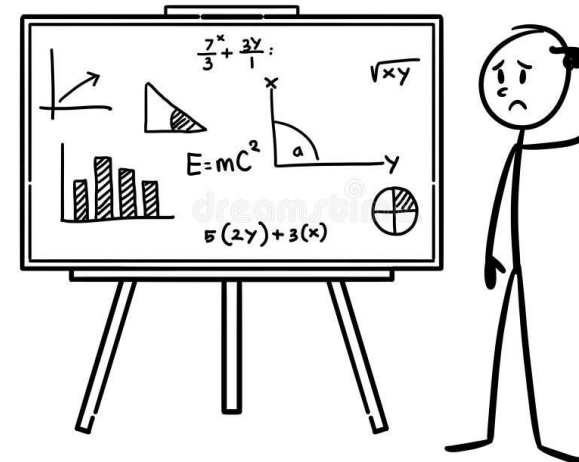
To improve AI's common sense, researchers are training it using **more realistic datasets** and **testing it with comprehensive checklists** that include **diverse situations**. This helps AI understand context better and avoid making illogical decisions

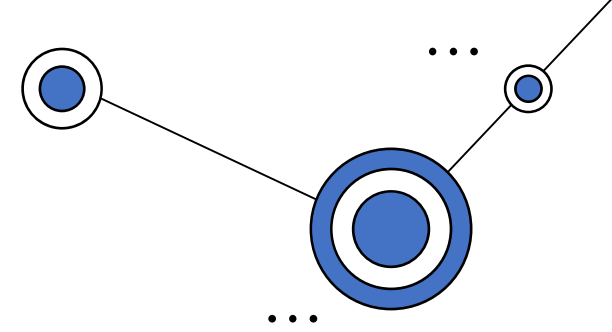




Math

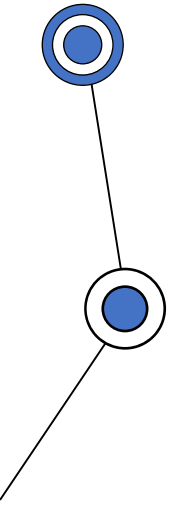
Despite their advanced capabilities, AI systems are surprisingly **bad at solving math problems**. Even state-of-the-art models struggle with **high school math**, achieving **low accuracy**. This is because AI **processes data in a parallel way**, while math requires **step-by-step reasoning**.





Remedies:

- Researchers are exploring **new AI architectures** that are better suited for **sequential problem-solving**. Providing AI with **more structured math datasets**.
- Teaching **AIs how to break down problems into smaller steps** can also help improve their performance.



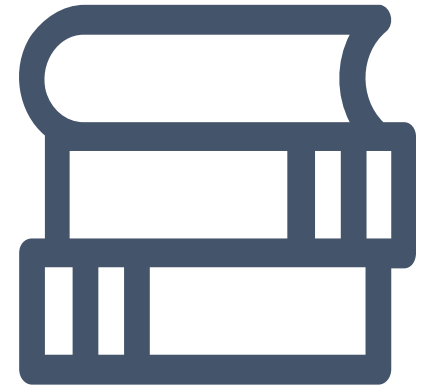
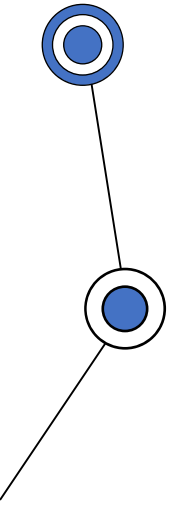
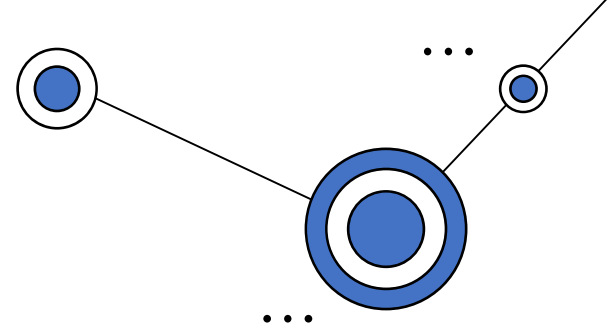
Conclusion

- AI has immense potential to improve our lives, but it's not without flaws. From **brittleness and bias to a lack of common sense and transparency**, these failures show us where AI needs improvement.
- Thankfully, researchers are working hard to address these challenges with creative solutions, like **better training methods, fairer algorithms, and smarter systems**.
- By understanding these **weaknesses** and their **remedies**, we can build AI that is not only more **reliable** but also **safer** and **fairer** for everyone.



Reference

- Choi, C. Q. (2021, October). *7 revealing ways AIs fail*. IEEE Spectrum. Retrieved from <https://spectrum.ieee.org>





**Thank You for
Your Attention**

The illustration features a person in a light blue shirt and dark pants, standing to the left of a large black-framed screen. The person is gesturing towards the screen with their right arm extended and their left hand raised. The screen displays the text 'Thank You for Your Attention' in a bold, black, serif font. The screen is set against a light blue, cloud-like background. The entire scene is framed by a network of blue lines and circular nodes, with some nodes containing concentric circles. The overall style is clean and modern, with a focus on the central message.